Intervals of validity

Up to now, we've been looking at various methods for solving specific types of differential equations, like linear, separable, Bernoulli, homogeneous, and exact equations.

We also learned to how to use an initial condition to solve an initial value problem, generating a particular solution from the differential equation's general solution. In this lesson we want to look at more initial value problems, and also address an important point that we've so far left out of our discussion.

Interval of validity for an initial condition

Now that we understand the idea of a solution curve through a direction field representing the solution to the differential equation, we also have to talk about the interval on which the solution is valid. We call this the **interval of validity**, interval of existence, interval of definition, or the domain of the solution.

An interval of validity can be an open interval like (a, b), a closed interval like $[a, b]$, or an infinite interval like $(-\infty, \infty)$ or (a, ∞).

As an example, we looked earlier at the linear differential equation

$$\frac{dy}{dx} + 2y = 4e^{-2x}$$

and used the integrating factor method to find its general solution

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$$y = \frac{4x + C}{e^{2x}}$$

Given the initial condition $y(0) = 1$, we could determine that the associated particular solution is

$$y = \frac{4x + 1}{e^{2x}}$$

and the interval of validity would be all real numbers, $(-\infty, \infty)$. Later in this lesson we'll explain in depth why this is the interval of validity, and exactly how to find intervals of validity for linear and non-linear differential equations.

But for now, we just want to know that we state an interval of validity that's associated with a specific initial condition, and therefore with a specific particular solution to the differential equation.

Linear and non-linear equations

Our process for finding the interval of validity will depend on whether we're starting with a linear differential equation or a non-linear differential equation.

When the equation is linear, we can actually find the interval of validity without calculating the equation's solution, because the interval of validity is determined by $P(x)$ and $Q(x)$ from the standard form of the linear equation,

$$\frac{dy}{dx} + P(x)y = Q(x)$$

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If $P(x)$ and/or $Q(x)$ are discontinuous at any point x, then those values of x will break up $(-\infty, \infty)$ into different potential intervals of validity. For instance, if $P(x)$ is discontinuous at $x = 0$ and $Q(x)$ is discontinuous at $x = 0$ and $x = \pm 3$, then we'd divide $(-\infty, \infty)$ into four distinct intervals,

$$(-\infty, -3) \qquad (-3,0) \qquad (0,3) \qquad (3,\infty)$$

As a rule, if we find n points of discontinuity among $P(x)$ and/or $Q(x)$, we'll divide $(-\infty, \infty)$ into $n + 1$ intervals. Of the $n + 1$ intervals, the correct interval of validity will be the one that contains t_0 from the given initial condition. So if we were given the condition $y(2) = -3$ and the intervals above, because $t_0 = 2$ falls within $(0,3)$, the interval $(0,3)$ is the interval of validity for the condition $y(2) = -3$ and the associated particular solution.

Notice how we define the interval of validity based on the given initial condition. The solution curve through $y(2) = -3$ had an interval of validity $(0,3)$, but a solution curve through $y(-1) = 0$ would have the interval of validity $(-3,0)$.

Let's do another example where we work through a full initial value problem for a linear differential equation.

Example

Find the interval of validity for the solution to the differential equation, given $f(0) = 5/2$.

$$\frac{dy}{dx} = -5y + 3e^x$$

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Put the equation in standard form of a linear equation.

$$\frac{dy}{dx} + 5y = 3e^x$$

From this new form, we can identify $P(x) = 5$ and $Q(x) = 3e^x$. Both of these functions are defined everywhere $(-\infty, \infty)$, so the interval of validity is $(-\infty, \infty)$.

If we wanted to also find the particular solution by solving the initial value problem, we'd find the integrating factor,

$$\rho(x) = e^{\int P(x)\, dx}$$

$$\rho(x) = e^{\int 5\, dx}$$

$$\rho(x) = e^{5x}$$

multiply through the linear equation,

$$\frac{dy}{dx}(e^{5x}) + 5y(e^{5x}) = 3e^x(e^{5x})$$

$$\frac{dy}{dx}e^{5x} + 5e^{5x}y = 3e^{6x}$$

$$\frac{d}{dx}(ye^{5x}) = 3e^{6x}$$

integrate both sides,

$$\int \frac{d}{dx}(ye^{5x})\, dx = \int 3e^{6x}\, dx$$

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$$ye^{5x} = \frac{1}{2}e^{6x} + C$$

and then solve for y to find the general solution.

$$y = \frac{e^{6x} + 2C}{2e^{5x}}$$

Substituting the initial condition $f(0) = 5/2$ lets us solve for C,

$$\frac{5}{2} = \frac{e^{6(0)} + 2C}{2e^{5(0)}}$$

$$\frac{5}{2} = \frac{1 + 2C}{2(1)}$$

$$5 = 1 + 2C$$

$$4 = 2C$$

$$C = 2$$

which we can plug back into the general solution to find the particular solution.

$$y = \frac{e^{6x} + 2(2)}{2e^{5x}}$$

$$y = \frac{e^{6x} + 4}{2e^{5x}}$$

This example illustrated how we were able to determine the interval of validity for the linear equation, without first finding the solution to the

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differential equation. This won't be the case when the equation is non-linear. Given a non-linear differential equation, we have to find the solution first, then determine where (if anywhere) the solution is undefined. The value(s) at which the solution is undefined will dictate the interval of validity.

Let's do an example with a separable differential equation. Previously, in the lesson about separable equations, we found the general solution

$$y = \frac{1}{\cos x + C}$$

to the equation $y' = y^2 \sin x$. We'll start with this general solution.

Example

Given the general solution to $y' = y^2 \sin x$ and the initial condition $y(0) = 2$, find the particular solution and the interval of validity.

$$y = \frac{1}{\cos x + C}$$

To find the particular solution, we'll substitute $x = 0$ and $y = 2$,

$$2 = \frac{1}{\cos 0 + C}$$

$$2 = \frac{1}{1 + C}$$

$$2 + 2C = 1$$

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$$2C = -1$$

$$C = -\frac{1}{2}$$

and then plug this back into the general solution.

$$y = \frac{1}{\cos x - \frac{1}{2}}$$

$$y = \frac{1}{\frac{-1 + 2\cos x}{2}}$$

$$y = \frac{2}{-1 + 2\cos x}$$

The solution is undefined where the denominator is 0.

$$-1 + 2\cos x = 0$$

$$2\cos x = 1$$

$$\cos x = \frac{1}{2}$$

$$x = \left\{ \frac{\pi}{3} + 2\pi k, \;\; k \in \mathbb{Z} \right\} \cup \left\{ \frac{5\pi}{3} + 2\pi k, \;\; k \in \mathbb{Z} \right\}$$

These values separate $(-\infty, \infty)$ into infinitely many intervals of validity.

$$\ldots\left(-\frac{7\pi}{3}, -\frac{5\pi}{3}\right) \quad \left(-\frac{5\pi}{3}, -\frac{\pi}{3}\right) \quad \left(-\frac{\pi}{3}, \frac{\pi}{3}\right) \quad \left(\frac{\pi}{3}, \frac{5\pi}{3}\right) \quad \left(\frac{5\pi}{3}, \frac{7\pi}{3}\right)\ldots$$

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Because the initial condition is $y(0) = 2$, and the value $x_0 = 0$ is contained in $(-\pi/3, \pi/3)$, we can say that $(-\pi/3, \pi/3)$ is the interval of validity.

When the interval of validity depends on y_0

Intervals of validity for non-linear equations can depend on the value of y_0 from the initial condition $y(t_0) = y_0$, whereas intervals of validity for linear equations are only dependent on t_0.

Let's do an example to see why this is the case.

Example

Find the interval of validity for the solution to the differential equation, given $y(0) = y_0$.

$$y' = y^2$$

This is a separable differential equation, which means it's non-linear, and therefore we'll need to find the solution to the equation before we can find the interval of validity. We'll first find the general solution,

$$\frac{dy}{dx} = y^2$$

$$dy = y^2 \, dx$$

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$$\frac{1}{y^2}\,dy = dx$$

$$\int \frac{1}{y^2}\,dy = \int dx$$

$$-\frac{1}{y} = x + C$$

$$-1 = (x + C)y$$

$$y = -\frac{1}{x + C}$$

and then we'll substitute the initial condition $y(0) = y_0$ to find a value for C.

$$y_0 = -\frac{1}{0 + C}$$

$$Cy_0 = -1$$

$$C = -\frac{1}{y_0}$$

Now we can plug this value back into the general solution, which will give us the equation's particular solution.

$$y = -\frac{1}{x - \frac{1}{y_0}}$$

$$y = -\frac{1}{\frac{xy_0}{y_0} - \frac{1}{y_0}}$$

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$$y = -\frac{1}{\frac{xy_0 - 1}{y_0}}$$

$$y = -\frac{y_0}{xy_0 - 1}$$

$$y = \frac{y_0}{1 - xy_0}$$

This particular solution is undefined where the denominator is 0.

$$1 - xy_0 = 0$$

$$1 = xy_0$$

$$x = \frac{1}{y_0}$$

Because the solution is undefined at this value, it divides $(-\infty, \infty)$ into two potential intervals of validity.

$$-\infty < x < \frac{1}{y_0} \qquad \frac{1}{y_0} < x < \infty$$

Since we're using the initial condition $y(0) = y_0$, the interval of validity must contain $x = 0$. The first interval will only contain $x = 0$ when

$$\frac{1}{y_0} > 0$$

$$y_0 > 0$$

The second interval will only contain $x = 0$ when

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$$\frac{1}{y_0} < 0$$

$$y_0 < 0$$

So we've found constraints for $y_0 < 0$ and $y_0 > 0$, but what about $y_0 = 0$? If $y_0 = 0$, then

$$y = \frac{y_0}{1 - xy_0} = \frac{0}{1 - x(0)} = 0$$

so $y = 0$ for all x, and the interval of validity is $(-\infty, \infty)$. So we can summarize our findings as

For $y_0 < 0$,	the interval of validity is	$\frac{1}{y_0} < x < \infty$
For $y_0 = 0$,	the interval of validity is	$(-\infty, \infty)$
For $y_0 > 0$,	the interval of validity is	$-\infty < x < \frac{1}{y_0}$

So we see that, even for a single initial condition, $y(0) = y_0$, the interval of validity will depend on the value of y_0. For instance, given the initial condition $y(0) = 1$, the interval of validity would be $(-\infty, 1)$. But given the initial condition $y(0) = -3$, the interval of validity would be $(-1/3, \infty)$.

In other words, we're finding different intervals of validity for the same value $x = 0$ in the initial condition.

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